Exercise semester 1/2569

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Exercise 3 Cost management (function point, cocomo II, earned value analysis) (45 points)

Submit your answer in a **pdf file**. Name the file by **your name and student ID.**

(23 points) Function point

1. (4 points) A web admin uses data analytic software to collect interaction data per event (or hit such as pageview, button click) when it happens. Collected data in an event table include timestamp, user ID, session ID, page URL, event name etc. At the end of each week, the web admin displays a weekly report showing the activity level by day of the week, i.e. the number of hits, % of total hits, and the number of user sessions. Calculate the function point of this function of the software.

Activity Level by Day of the Week Display

Activity Level by Day of the Week

	Day	Hits	% of Total Hits	User Sessions
1	Sun	1004	8.73%	111
2	Mon	1887	16.41%	201
3	Tue	1547	13.45%	177
4	Wed	1975	17.17%	195
5	Thu	1591	13.83%	191
6	Fri	2209	19.21%	200
7	Sat	1286	11.18%	121
	Total Weekdays	**9209**	**80.08%**	**964**
	Total Weekend	**2290**	**19.91%**	**232**

Which function (EI, EO, EQ, ILF, or EIF)?	EO – the report sends derived data across the boundary to the user: the hits, % of total hits, user sessions and weekday/weekend totals are calculated from the event records, not just retrieved, so it is an External Output (not an EQ).
What is the number of DETs and what are they?	8 DETs: 1. Display button (input, action key) 2. Day (output, recursive – Sun ... Sat) 3. Number of hits (output, derived – count of events per day, recursive) 4. % of total hits (output, derived, recursive) 5. Number of user sessions (output, derived – count of sessions per day, recursive) 6. Total hits (weekdays / weekend) (output, derived) 7. Total % of hits (weekdays / weekend) (output, derived) 8. Total user sessions (weekdays / weekend) (output, derived) (The row number 1–7 and the report title are static labels, not DETs.)

	(continuation of the DET cell – all 8 DETs are listed on page 1)
If counting FTRs is applicable to this function, what is the number of FTRs and what are they? If not, write N/A.	1 FTR: 1. Event table (ILF – the interaction data collected per hit, maintained by the software)
If counting RETs is applicable to this function, what is the number of RETs and what are they? If not, write N/A.	N/A – RETs are counted only for data functions (ILF, EIF), not for an EO.
Rating (Low, Average, or High)?	Low (EO with 1 FTR [FTR band 0–1] and 8 DETs [DET band 6–19] – slide 37 table)
Value?	**4 (EO, Low = 4 function points)**

2. (4 points) The US Census Bureau hosts a web application that allows users to click on a location on the map (e.g. map location of Kansas City) to see some information about that location (i.e. city, state, population, and state coordinate). The US Census Bureau maintains its own state population data and obtains the state coordinate data from the Surveying and Mapping Section of the US Army [กรมแผนที่ทหาร]. Calculate the function point of this function of the application.



Kansas City, Missouri: Population 435,146: Location: 39.1 N, 94.5 W

Which function (EI, EO, EQ, ILF, or EIF)?	EQ – input (the click on a map location) plus output that only retrieves stored data (city, state, population, coordinate) from the files; nothing is derived and no ILF is updated, so it is an External Inquiry.
What is the number of DETs and what are they?	5 DETs: 1. Map location clicked (input, action key – e.g. Kansas City) 2. City (output) 3. State (output) 4. Population (output) 5. State coordinate (output – e.g. 39.1 N, 94.5 W)
If counting FTRs is applicable to this function, what is the number of FTRs and what are they? If not, write N/A.	2 FTRs: 1. State population data (ILF – maintained by the US Census Bureau) 2. State coordinate data (EIF – maintained by the Surveying and Mapping Section of the US Army, only referenced by this application)
If counting RETs is applicable to this function, what is the number of RETs and what are they? If not, write N/A.	N/A – RETs are counted only for data functions (ILF, EIF), not for an EQ.
Rating (Low, Average, or High)?	Low (EQ with 2 FTRs [FTR band 2–3] and 5 DETs [DET band 1–5] – slide 37 table)
Value?	**3 (EQ, Low = 3 function points)**

3. (4 points) The Mor Prom [หมอพร้อม] application of the Ministry of Public Health (MOPH) is the the government's health application, first launched to provide the Covid vaccination service. Users provide personal data to register with this application, and the data are kept as part of the user data. Calculate the function point of this function of the application.





Which function (EI, EO, EQ, ILF, or EIF)?	EI – the user's personal data cross the boundary from outside (the user) into the application and maintain an ILF (the user data): the Mor Prom registration form is an External Input.
What is the number of DETs and what are they?	18 DETs: 1. Title (คำนำหน้าชื่อ, drop-down list) 2. First name (ชื่อ) 3. Last name (นามสกุล) 4. National ID number (เลขประจำตัวประชาชน) 5. Birth day (วันที่เกิด, drop-down list) 6. Birth month (เดือนที่เกิด, drop-down list) 7. Birth year (ปีที่เกิด, drop-down list) 8. Phone number (เบอร์โทรศัพท์) 9. Gender (เพศ – เพศชาย/เพศหญิง, one DET for the pair of radio-style buttons) 10. Address (ที่อยู่) 11. Province (จังหวัด, drop-down list) 12. District (อำเภอ, drop-down list) 13. Sub-district (ตำบล, drop-down list) 14. Postcode (รหัสไปรษณีย์) 15. Hospital (เลือกโรงพยาบาล, drop-down list) 16. Save button (บันทึก – the action key that saves the registration) 17. Error message (e.g. “กรุณากรอกข้อมูลให้ครบ” / invalid national ID) 18. Confirmation message (registration successful) (15 personal data fields, as question 4 states, + 1 action key + 1 error message + 1 confirmation message. The greyed-out target group (กลุ่มเป้าหมาย = ประชาชนทั่วไป) is filled by the system, not entered by the user, so it is not counted; counting it (19 DETs) would not change the rating. The drop-down choices for province/district/hospital are code data, not logical files, so they add no FTR.)
If counting FTRs is applicable to this function, what is the number of FTRs and what are they? If not, write N/A.	1 FTR: 1. User data (ILF – maintained by the registration, i.e. the personal data are stored in it)
If counting RETs is applicable to this function, what is the number of RETs and what are they? If not, write N/A.	N/A – RETs are counted only for data functions (ILF, EIF), not for an EI.
Rating (Low, Average, or High)?	Average (EI with 1 FTR [FTR band 0–1] and 18 DETs [DET band > 15] – slide 37 table)
Value?	**4 (EI, Average = 4 function points)**

4. (4 points) The Mor Prom [หมอพร้อม] application stores the user data which consist of the user's personal data, i.e. 15 data fields, and the information about vaccine shots that the user has received, i.e. date of vaccination, vaccine name, vaccine manufacturer, vaccine lot, and vaccination site. Calculate the function point of this function of the application.

Which function (EI, EO, EQ, ILF, or EIF)?	ILF – a user-identifiable group of logically related data (the user data) kept entirely inside the Mor Prom boundary and maintained by the application itself (registration, recording of shots), so it is an Internal Logical File.
What is the number of DETs and what are they?	20 DETs: 1. Personal data: 15 data fields (title, first name, last name, national ID, birth day, birth month, birth year, phone number, gender, address, province, district, sub-district, postcode, hospital) 2. Date of vaccination (vaccine shot data) 3. Vaccine name (vaccine shot data) 4. Vaccine manufacturer (vaccine shot data) 5. Vaccine lot (vaccine shot data) 6. Vaccination site (vaccine shot data)
If counting FTRs is applicable to this function, what is the number of FTRs and what are they? If not, write N/A.	N/A – FTRs are counted only for transactions (EI, EO, EQ), not for an ILF.
If counting RETs is applicable to this function, what is the number of RETs and what are they? If not, write N/A.	2 RETs: 1. Personal data (parent – the user) 2. Vaccine shot data (child – one user has many shots; a shot is meaningless without its user)
Rating (Low, Average, or High)?	Average (ILF with 2 RETs [RET band 2–5] and 20 DETs [DET band 20–50] – slide 37 table)
Value?	**10 (ILF, Average = 10 function points)**

5. (4 points) The Mor Prom [หมอพร้อม] application connects to the Organ Donation API of the Thai Red Cross to access users' organ donation data. The organ donation data are maintained by the Thai Red Cross system and are only referenced by the Mor Prom application. The organ donation data consist of donor information and information about the donated organs. The donor information includes national ID, title, first name, last name, date of birth, occupation, address, tumbol, district, province, postcode,

house phone, and mobile phone. For each donated organ, the information includes the donation date, donated organ, and witness name. Calculate the function point of this function of the application.

Which function (EI, EO, EQ, ILF, or EIF)?	EIF – a group of logically related data (the organ donation data) that Mor Prom only references; it lives outside the boundary and is maintained by another application (the Thai Red Cross system, via the Organ Donation API), so it is an External Interface File.
What is the number of DETs and what are they?	16 DETs: 1. National ID 2. Title 3. First name 4. Last name 5. Date of birth 6. Occupation 7. Address 8. Tumbol 9. District 10. Province 11. Postcode 12. House phone 13. Mobile phone 14. Donation date 15. Donated organ 16. Witness name (13 donor-information fields + 3 fields per donated organ.)
If counting FTRs is applicable to this function, what is the number of FTRs and what are they? If not, write N/A.	N/A – FTRs are counted only for transactions (EI, EO, EQ), not for an EIF.
If counting RETs is applicable to this function, what is the number of RETs and what are they? If not, write N/A.	2 RETs: 1. Donor information (parent – the donor) 2. Donated organ information (child – one donor may donate several organs: donation date, donated organ, witness name)
Rating (Low, Average, or High)?	Low (EIF with 2 RETs [RET band 2–5] and 16 DETs [DET band 1–19] – slide 37 table)

Value?	**5 (EIF, Low = 5 function points)**

6. (3 points) Suppose an application has the following number of functions of various sizes.

	No. of low	No. of average	No. of high
External Input (EI)	3	1	0
External Output (EO)	1	0	0
External Inquiry (EQ)	2	1	0
Internal Logical File (ILF)	5	0	0
External Interface File (EIF)	1	0	0

Calculate the unadjusted function points of this application.

	Low	Average	High	Total
External Input (EI)	3 *3 = 9	1 *4 = 4	0 *6 = 0	**13**
External Output (EO)	1 *4 = 4	0 *5 = 0	0 *7 = 0	**4**
External Inquiry (EQ)	2 *3 = 6	1 *4 = 4	0 *6 = 0	**10**
Internal Logical File (ILF)	5 *7 = 35	0 *10 = 0	0 *15 = 0	**35**
External Interface File (EIF)	1 *5 = 5	0 *7 = 0	0 *10 = 0	**5**
			Unadjusted FP	**67**

Unadjusted FP = sum of (count × weight) = EI (3×3 + 1×4 + 0×6 = 13) + EO (1×4 = 4) + EQ (2×3 + 1×4 = 10) + ILF (5×7 = 35) + EIF (1×5 = 5) = 13 + 4 + 10 + 35 + 5 = 67 function points (weights from slide 43).

(8 points) COCOMO II Post-architecture Model

1. Suppose the application size is 67 function points. If the application is written in Javascript and the average gearing factor is **47**....., the application size is **67 × 47 / 1000 = 3.149**..... KLOC. (See gearing factor here https://www.qsm.com/resources/function-point-languages-table)

(QSM Function Point Languages Table v5.0: JavaScript gearing factor avg 47, median 53, range 31–63 SLOC/FP; KLOC = FP × gearing factor / 1000, slide 47.)

2. Scale Factor (SF) (Rate SF as you see fit.)

Scale Factor (SF)	Rating	Score
Precedentedness (PREC)	High	2.48
Development Flexibility (FLEX)	High	2.03
Architecture / Risk Resolution (RESL)	Nominal	4.24
Team Cohesion (TEAM)	High	2.19
Process Maturity (PMAT)	Nominal	4.68
sum of SF = 2.48 + 2.03 + 4.24 + 2.19 + 4.68 (values from slide 54, Table 10)	Σ SF	**15.62**

3. Effort Multipliers (EM) (Rate EM as you see fit.)

Effort Multiplier (EM)	Rating	Score
Product factors		
Required Software Reliability (RELY)	Nominal	1.00
Data Base Size (DATA)	Low	0.90
Product Complexity (CPLX)	Low	0.87
Developed for Reusability (RUSE)	Nominal	1.00
Documentation Match to Life-Cycle Needs (DOCU)	Nominal	1.00
Platform factors		
Execution Time Constraint (TIME)	Nominal	1.00
Main Storage Constraint (STOR)	Nominal	1.00
Platform Volatility (PVOL)	Low	0.87
Personnel factors		
Analyst Capability (ACAP)	Nominal	1.00
Programmer Capability (PCAP)	Nominal	1.00
Personnel Continuity (PCON)	Nominal	1.00
Applications Experience (APEX)	Low	1.10
Platform Experience (PLEX)	Low	1.09
Language and Tool Experience (LTEX)	Nominal	1.00
Project factors		

Use of Software Tools (TOOL)	Nominal	1.00
Multisite Development (SITE)	High	0.93
Required Development Schedule (SCED)	Low	1.14
product of EM = 0.90 × 0.87 × 0.87 × 1.10 × 1.09 × 0.93 × 1.14 (all other EMs = 1.00) = 0.8659	$\prod$ EM	**0.8659**

Show <u>how you calculate</u> for the following. (แทนค่าในสมการให้ดูด้วย)

4. E = B + 0.01 × (sum of SF) = 0.91 + 0.01 × 15.62 = 1.0662 .....

5. Project effort PM = A × Size^E × (product of EM) = 2.94 × 3.149^1.0662 × 0.8659 = 8.65 ..... person-months

6. Nominal effort PM_{NS} = 2.94 × 3.149^1.0662 × 0.7596 (EM product without SCED) = 7.59 ..... person-months

7. Calendar time to develop TDEV is 3.67 × 7.59^0.3112 × 85/100 = 5.86 ..... months

8. Staff size required to do the amount of work in 5. during the calendar time in 7. is PM / TDEV = 8.65 / 5.86 = 1.48, i.e. 2 persons (rounded up) ..... persons

Calculations (COCOMO II.2000 post-architecture model, slides 52, 66 and 67; A = 2.94, B = 0.91, C = 3.67, D = 0.28; Size = 3.149 KLOC from step 1):

4. E = B + 0.01 × (sum of SF) = 0.91 + 0.01 × (2.48 + 2.03 + 4.24 + 2.19 + 4.68) = 0.91 + 0.01 × 15.62 = 0.91 + 0.1562 = 1.0662

5. PM = A × Size^E × (product of EM) = 2.94 × 3.149^1.0662 × (1.00 × 0.90 × 0.87 × 1.00 × 1.00 × 1.00 × 1.00 × 0.87 × 1.00 × 1.00 × 1.00 × 1.10 × 1.09 × 1.00 × 1.00 × 0.93 × 1.14)
= 2.94 × 3.3974 × 0.8659 = 8.65 person-months

6. PM_NS = effort with a nominal schedule, i.e. the same product without the SCED multiplier (1.14):
PM_NS = 2.94 × 3.149^1.0662 × (0.8659 / 1.14) = 2.94 × 3.3974 × 0.7596 = 7.59 person-months

7. TDEV = [C × PM_NS^(D + 0.2 × (E – B))] × SCED% / 100
= [3.67 × 7.59^(0.28 + 0.2 × (1.0662 – 0.91))] × 85 / 100 = [3.67 × 7.59^0.3112] × 0.85
= [3.67 × 1.8790] × 0.85 = 6.90 × 0.85 = 5.86 months (SCED rated Low = 85% of the nominal schedule)

8. Staff size = PM / TDEV = 8.65 person-months / 5.86 months = 1.48 persons, i.e. about 1.5 full-time persons – in practice 2 persons (rounded up; with 2 persons the work takes 4.3 months, within the 5.86-month schedule).

Rating rationale (steps 2 and 3) – a small JavaScript web application built by a small co-located team:
- PREC High (2.48): generally familiar – a web application like many built before
- FLEX High (2.03): general conformity – few pre-established external interface constraints
- RESL Nominal (4.24): about 60% of risk items identified and resolved, architecture reviewed
- TEAM High (2.19): largely cooperative stakeholders and team
- PMAT Nominal (4.68): SW-CMM level 2 – basic project planning and tracking
- RELY Nominal (1.00): moderate, easily recoverable losses if it fails
- DATA Low (0.90): small testing database, D/P < 10
- CPLX Low (0.87): simple control, data and user-interface operations
- RUSE Nominal (1.00): reusable across the project only
- DOCU Nominal (1.00): documentation right-sized to life-cycle needs
- TIME Nominal (1.00): ≤ 50% use of available execution time
- STOR Nominal (1.00): ≤ 50% use of available storage
- PVOL Low (0.87): stable platform – major change every 12 months
- ACAP Nominal (1.00): analysts at the 55th percentile
- PCAP Nominal (1.00): programmers at the 55th percentile
- PCON Nominal (1.00): 12% annual turnover
- APEX Low (1.10): about 6 months of experience with this kind of application
- PLEX Low (1.09): about 6 months of experience with the web platform
- LTEX Nominal (1.00): about 1 year of JavaScript and tool experience
- TOOL Nominal (1.00): basic life-cycle tools, moderately integrated
- SITE High (0.93): same city, wideband electronic communication
- SCED Low (1.14): schedule compressed to 85% of nominal

(6 points) Cost estimation

Show your calculations for the following.

Suppose the estimated effort for the project is 5 person-months. The project has a fixed number of staff that will be working on the project and agile methodology (Scrum) is used. The distribution of the project effort on different project activities and on different sprint activities is shown in the table below.

Project Activity	**% of Project Effort PM**
Management	10%
Analysis	10%
Sprint 1	20%
Sprint 2	20%
Sprint 3	20%
Sprint 4	20%

Sprint Activity	**% of Sprint Effort**
Analysis	5
Design	15
Implementation	50
Testing	20
Management	10

Sprint Activity	**% of Sprint Effort**
Analysis	5
Design	15
Implementation	50
Testing	20
Management	10

Sprint Activity	**% of Sprint Effort**
Analysis	5
Design	15
Implementation	50
Testing	20
Management	10

Sprint Activity	**% of Sprint Effort**
Analysis	5
Design	15
Implementation	50
Testing	20
Management	10

In the table below, show how to calculate % of project effort of each project activity, activity effort in person-months, activity labor cost, and total project labor cost. **(แสดงการคำนวณให้ดูด้วย)**

Project Activity	% of Project Effort PM	Activity Effort (Person-Months)	Average Activity Labor Cost Per Person-Month (THB)	Activity Labor Cost (THB)
Management	10% + 4×20%×10% = 18%	18% × 5 = 0.9	90,000	0.9 × 90,000 = 81,000
Analysis	10% + 4×20%×5% = 14%	14% × 5 = 0.7	50,000	0.7 × 50,000 = 35,000
Design	4×20%×15% = 12%	12% × 5 = 0.6	50,000	0.6 × 50,000 = 30,000
Implementation	4×20%×50% = 40%	40% × 5 = 2	30,000	2 × 30,000 = 60,000
Testing	4×20%×20% = 16%	16% × 5 = 0.8	30,000	0.8 × 30,000 = 24,000
			Total Project Labor Cost	**230,000**

Calculations (effort 5 person-months; management 10% and analysis 10% are project-level activities, the remaining 80% is spread over 4 sprints of 20% each, and inside every sprint the effort is split into analysis 5%, design 15%, implementation 50%, testing 20% and management 10%):

% of project effort of an activity = its project-level % + the sum over the 4 sprints of (sprint % of project × activity % of sprint):
- Management: 10% + 4 × (20% × 10%) = 10% + 4 × 2% = 10% + 8% = 18%
- Analysis: 10% + 4 × (20% × 5%) = 10% + 4 × 1% = 10% + 4% = 14%
- Design: 4 × (20% × 15%) = 4 × 3% = 12% = 12%
- Implementation: 4 × (20% × 50%) = 4 × 10% = 40% = 40%
- Testing: 4 × (20% × 20%) = 4 × 4% = 16% = 16%

  Check: 18% + 14% + 12% + 40% + 16% = 100% of the project effort.

Activity effort (person-months) = % of project effort × 5 PM; activity labor cost = activity effort × average labor cost per person-month:
- Management: 18% × 5 = 0.9 PM; 0.9 × 90,000 = 81,000 THB
- Analysis: 14% × 5 = 0.7 PM; 0.7 × 50,000 = 35,000 THB
- Design: 12% × 5 = 0.6 PM; 0.6 × 50,000 = 30,000 THB
- Implementation: 40% × 5 = 2 PM; 2 × 30,000 = 60,000 THB
- Testing: 16% × 5 = 0.8 PM; 0.8 × 30,000 = 24,000 THB

  Check: 0.9 + 0.7 + 0.6 + 2.0 + 0.8 = 5 PM.

Total project labor cost = 81,000 + 35,000 + 30,000 + 60,000 + 24,000 = 230,000 THB (an average of 46,000 THB per person-month).

(8 points) Earned Value Analysis

Assume we are through the second month of a one-year project that has a total budget of 1,200,000 bahts. The amount budgeted for the first two months is 230,000 bahts, but the work that has been finished in these two months is worth 200,000 bahts of the budget. The actual cost for the first two months is 250,000 bahts. **(แสดงการคำนวณให้ดูด้วย)**

Budget at completion (BAC) = 1,200,000 baht (the original total budget of the project)

Planned Value (PV) = 230,000 baht (the budget for the work scheduled in the first two months)

Earned Value (EV) = 200,000 baht (the budgeted value of the work actually finished)

Actual Cost (AC) = 250,000 baht (the actual cost of the first two months)

Cost Variance (CV) = EV – AC = 200,000 – 250,000 = –50,000 baht (negative: over budget)

Schedule Variance (SV) = EV – PV = 200,000 – 230,000 = –30,000 baht (negative: behind schedule)

Cost Performance Index (CPI) = EV / AC = 200,000 / 250,000 = 0.80 (80%: 20% over budget)

Schedule Performance Index (SPI) = EV / PV = 200,000 / 230,000 = 0.8696 (86.96%: 13% behind schedule)

Estimate at completion (EAC) = BAC / CPI = 1,200,000 / 0.80 = 1,500,000 baht

Estimated time to complete = original time estimate / SPI = 12 months / 0.8696 = 13.80 months

Estimated to complete (ETC) = EAC – AC = 1,500,000 – 250,000 = 1,250,000 baht

To complete the project with the original budget, To-complete performance index (TCPI) = (BAC – EV) / (BAC – AC) = (1,200,000 – 200,000) / (1,200,000 – 250,000) = 1,000,000 / 950,000 = 1.0526

1. Considering the Cost Performance Index (CPI) and Schedule Performance Index (SPI), how is the project doing?

☐ under budget ☐ on budget ☑ over budget (CPI = 0.80 < 1)

☐ ahead of schedule ☐ on schedule ☑ behind schedule (SPI = 0.87 < 1)

2. If the team continues to perform at the current cost efficiency, the cost of completing the project would be EAC = BAC / CPI = 1,500,000 baht (300,000 baht or 25% more than the BAC)

3. If the team continues to perform at the current schedule efficiency, the time of completing the project would be 12 / SPI = 13.80 months (about 1.8 months or 15% longer than planned)

4. Considering the To-complete performance index (TCPI), is the original budget reasonable for completing the project? Why? Yes, still reasonable, but only with better cost efficiency: TCPI = 1.05 > 1: the remaining 1,000,000 baht of work must be done with the remaining 950,000 baht, so the cost efficiency must rise from CPI 0.80 to 1.05 (+32%). TCPI – CPI = 0.25 is within 0.5 × CPI = 0.40, so (slide 79) the BAC is reasonable.

(The required improvement is demanding but not overly optimistic by the rule of thumb; if the team keeps CPI = 0.80, the project would cost EAC = 1,500,000 baht instead of the 1,200,000-baht budget.)